Drone-aided Detection of Weeds: Transfer Learning for Embedded Image Processing

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***Abstract*—In this article, we address the problem of Hogweed detection using a drone equipped with an RGB and multispectral cameras. We study two approaches (i) offline detection running on the orthophoto of the area scanned within the mission, and (ii) realtime scanning from the frames stream directly on the edge device performing the flight mission. We show that by fusing the information from an additional multispectral camera installed on the drone, there is an opportunity to boost the detection quality which then can be preserved even with a single RGB camera setup by introduction of additional Convolution Neural Network (CNN) trained with transfer learning to produce the fake multispectral images directly from the RGB stream. We show that this approach helps either eliminate the multispectral hardware from the drone or if only the RGB camera is at hand, to boost the segmentation performance by the cost of slight increase in computational budget. To support this claim we have performed an extensive study of networks performance in simulations of both realtime and offline modes, where we achieve at least 1.1% increase in terms of mean intersection over union (mIoU) metric when evaluated on RGB stream from the camera and 1.4% when evaluated on ortophoto data. Our results show that the proper optimization guarantees a complete elimination of the multispectral camera from the flight mission by adding a preprocessing stage to segmentation network without the loss of quality.**

***Index Terms*—Semantic segmentation, deep learning, transfer learning, edge computing**

I. Introduction

CONTINUOUSLY growing human population imposes strict demands to agricultural industry in terms of improving the crop yields and making the food production efficient. At the same time, high efficiency can be achieved by controlling and removing the weeds. To overcome this challenge a high number of the state-of-the-art technologies have been applied in the scope of precision agriculture [1].

Distribution of weeds and, in particular, Hogweed of Sosnowsky, is a quickly growing problem for agricultural industry. Hogweed of Sosnowsky is a weed that has spread across Eurasia region, including European and Asian countries, Great Britain and Russia, as well as local parts of North American continent. Hogweed typically grows for up to 5 *m* and may have the diameter for up to 12 *cm*. It is capable of producing thousands of seeds which are then distributed by wind across large area. The plant produces toxins which are hazardous for human, but at the same time hogweed creates a danger to other crops including the farming ones by affecting their ecosystem. This hazardous weed must, therefore, be precisely identified and removed.

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Traditionally, there are three typical approaches for weeds localization: (i) inspection by human, (ii) satellite imaging, and (iii) Unmanned Aerial Vehicles (UAV) monitoring. We discuss these approaches in Section II.

In this work, we rely on the UAV option due to its practical feasibility. The most commonly utilized sensor setup for this platform consists of a single video-camera, capable of producing high-quality RGB images and a Global Navigation Satellite System sensor providing geographic coordinates for the measurements from the camera. Additional multispectral camera is also typically introduced for the crops monitoring missions to allow for a computation of vegetation indices. The same sensor could also be utilized to increase the crops detection accuracy as it was shown for example in [2], but exploiting two video cameras on board is not weight and cost wise.

In this paper we investigate the trade offs between semantic segmentation quality of crops and the imaging sensors setup complexity. We use a task of hogweed detection as a playground and propose a solution based on the advances of deep learning, which allows for the complete avoidance of a multispectral camera from the mission without any detoriation of segmentation performance compared to the multiple cameras platform. More specifically, we report on the following novelty:

1) Dataset collection, alignment and labeling. We propose to label the hogweed plants by transforming a sequence of overlapping frames into a single orthophoto. It enables labeling the crops in the area of interest only once without repetition in intersecting parts of the images. The corresponding labels are then back-transformed to the camera frames using a greedy features-based image alignment procedure. Similar procedure is applied to align the RGB and multispectral data, with the corresponding orthophotos acting as a reference. As a result our approach the labeled and aligned RGB and multispectral images are produced fast without a need to synchronize frame streams coming from two separate camera sensors. To the best of our knowledge, such methodology was never reported before.
2) Removal of multispectral camera from the setup by replacement with the preprocessing network. Instead of using two cameras setup, we propose to adopt a transfer learning procedure and learn a convolution neural network from the collected paired data to mimic multispectral images. Such images are then synthesised during preprocessing stage, which allows to completely avoid multispectral hardware in the aerial platform without any

deterioration of segmentation quality.

This article is organized as follows: we introduce the readers to the relevant works in the area in Section II. In Section III we discuss the methods used in this research. Our results are provided in Section IV. Finally, we provide concluding remarks in Section V.

II. Related Work

A. Weed Detection

Traditional approaches of weeds localization include manual inspection with special handheld equipment, which is done either by foot or with utilization of transport machines like cars or tractors [3]. Although these methods can still be in favour for farmers, there is a certain shift towards the systems allowing for monitoring of larger areas while preventing the spread of seeds by feet and wheels. That is why the modern development of methods involving the Unmanned Aerial Vehicles (UAV) and even satellites imaging starts to gain close attention for the purposes of more effective human-free ways of crops monitoring and weed control [4].

Currently satellite imaging provides accurate images across huge areas with different spectral bands lying far beyond just conventional visual spectrum, provided by ordinary RGB camera. However, satellite imaging still does not allow to perform precise crops localization due to the large Ground Sampling Distance (GSD), since resolution of the best available satellite images does not exceed 30 *cm/px*. Typical size of the hogweed leafs is around the same 30 *cm*, which results in a situation, when a single hogweed plant is resolved into image area of several pixels, which makes its detection nearly impossible. Also satellite monitoring system is vulnerable to the weather changes, since it becomes impossible to capture plants at a cloudy weather. At the same time, UAVs enable to select GSD from millimeters to meters per pixel depending on the needs by varying altitude during the mission. As for UAVs it is still impossible to perform monitoring in rainy conditions, the applicability of them is far more flexible, than satellites. Currently, the limitation on equipment and computational resources the UAVs can carry, as well as the area UAVs can cover per a single mission, are strongly bounded by their lifting capacity and power supply resources. This issue poses the problem of effective management of sensors and computer vision algorithms used for monitoring.

Nowadays in precision agriculture a multisensor setup is typically used. It combines a traditional RGB sensor together with the infrared one. For the latter the Near Infrared band (NIR) is usually considered. Compared to a single RGB camera, the multispectral cameras enable crops monitoring with much greater precision as well as direct calculation of their vegetation characteristics. This is done at a cost of additional hardware carried on board during the mission, and additional efforts for accurate sensors synchronization and images alignment.

B. Image Analysis in Precision Agriculture

Even before the raise of deep learning, many algorithms were proposed to perform a high-level vision of the images. Most of them rely on handcrafted distinct features of the object of interest, and it was shown that this principle is suitable for the agriculture as well [5]. Now almost all state-of-the-art approaches involve the usage of CNNs and even transformers. Ronneberger *et al*. [6] were one of the pioneers who introduced a U-shaped neural architecture for the task of image segmentation, which is still applied in a real high-level vision tasks. This architecture was shown to produce state-of-the-art results in agricultural tasks as well and being capable to run on the edge device. For example, in [7] U-shaped architecture was deployed to the unmanned ground vehicle (UGV) in order to perform a robust estimate a pixel-wise labeling of the images into crop and weed. It was shown, that this setup outperforms previous state-of-the-art approaches and improves the accuracy of crop-weed classification without requiring a retraining of the model. In [8] the authors reported on a comparison between other segmentation CNNs which showed that U-Net performance is slightly lower than the competing SegNet and RefineNet with ResNet backbone. However, the computational complexity of the latter two models does not allow for deploying them to an embedded unit and perform the inference in real-time during the mission.

In our previous work [8] we focused on the optimization of the most relevant FCNN architectures for inference on board the Single Board Computer (SBC). The point of this work was the feasibility of the application of such a drone with AI on board in Precision Agriculture (PA). However, the dataset was collected during a single day. It does not contain different phases of the hogweed growth. Thus, the capabilities of the FCNN, trained on such a dataset are limited. Furthermore, the data were collected using the RGB camera only. The current paper proposes an approach, that enhances the capabilities of the monitoring system in several ways.

First, the dataset used in this study was collected in the same area with a weekly period of 2.5 months. It contains the data on the growth phases of the hogweed during this timespan. Hence, such a dataset is valuable for both AI and PA areas. Moreover, the dataset was collected using both RGB and multispectral cameras. Additional channels give more information about the object of interest and improve the accuracy and robustness of the neural network. The current research's contribution is the enhancement of the neural network capabilities due to the generation of artificial NIR channels. It provides a significant advantage in terms of both accuracy of predictions and feasibility such a monitoring system in practice. Application of the proposed approach allows rid of the multispectral camera for hogweed detection. Thus it makes the monitoring system cheaper, more sustainable, and prolongs the flight time.

It should be noted that deployment of neural architecture to the edge device is not a trivial task [9] and various optimization techniques were proposed to overcome this problem. A hardware-based optimization for binary weight CNNs was proposed in [10] where the arithmetic core was optimized by elimination of the need for expensive multiplications, as well as by reduction of I/O bandwidth and storage. Typically, however, most well known techniques to accelerate inference are software-based, and include architecture-related optimizations which are performed either manually or automatically via

Neural Architecture Search (NAS) and pruning[1]. It was also shown that another type of optimization performed by quantizing neural network weights and inference procedure to uint8 or even binary can drastically accelerate the inference without large deterioration of their performance. Such techniques were used for example in [11] to accelerate semantic segmentation task on NVIDIA Jetson Xavier NX by the usage of computationally efficient manually designed adaptive blocks, namely multi-fiber unit and attention module, as well as quantization of model weights to int8 and pruning, helping achieve a good inference speed without a big drop of prediction quality.

C. Intelligent Monitoring Systems in Precision Agriculture

There are different types of autonomous vehicles, which allow a precise monitoring of crops. Typically they are classified into two groups, UAVs, that operate in the air, and UGVs, that operate on the ground. As UGVs are able to perform individual plant phenotyping, characteristics monitoring and even manipulations with crops and weeds [12], in crops monitoring tasks the priority is given to UAVs due to their ability to cover large areas relatively fast without disseminating the seeds.

Typically, the UAVs are equipped with an RGB camera capturing the crops underneath in a visual spectrum, and the machine vision algorithms are processing the images from this camera in order to get the desired output. Typically such processing is done *offline* after the mission is complete [13] since the frames stream is stored on an internal storage of UAV [14] or is transferred to the intermediate server during the mission by a wireless protocol [15]. Since both cases are not practically feasible as they require additional efforts and do not allow for fast decision making, it is important to consider inference being performed directly on the edge device [16]. This is especially vital for the hogweed detection, as the weed can spread very fast by generating up to 100,000 seeds a year which are quickly disseminated by wind. A proper detection and elimination of each poisonous plant is essential, and the edge computing enables it by providing the instant information on geolocation of the plant.

In order to improve the monitoring outcomes, additional information can be provided to the vision algorithm in a form of a separate multispectral frame stream. Typically this is done to allow direct calculations of vegetation indices such as normalized difference vegetation index (NDVI) or normalized difference red edge index (NDRE) [17]. For example, in [18] authors report a UAV-based setup consisting of RGB and NIR-retrofitted camera, which performs imaging missions over rice and corn fields in the Philippines. They comment on using dual camera setup in order to compute precise NDVI of the crops. A survey performed by Radoglou-Grammatikis *et al*. [19] extends the list of possible multispectral camera usages towards optimization of the image acquisition system and water stress indices. At the same time robust quantitative studies require a proper usage of multispectral cameras and additional attention not only to their temporal calibration with visual band cameras, but also to the calibration of their spectral characteristics, as it was highlighted in [20].

By this reason several approaches were proposed targeting capture of multispectral information directly from the corresponding RGB camera frames. For example, several approaches propose to remove the near infrared blocking filter installed inside all the consumer RGB cameras [21]. In this case the signal from the NIR band is mixed with the red channel, however authors claim that it still helps to calculate the NDVI with high tolerance. Costa *et al*. [22] introduce a new NDVI mimicking index computed from RGB data by the use of a genetic algorithm. Finally, inspired by Pix2Pix [23] performance in conditional synthesis tasks, Yuan *et al*. [24] and Soni *et al*. [25] separately proposed to use the same conditional generative adversarial CNN for the purpose of sampling visually plausible near infrared spectral band from the satellite and aerial RGB observations respectively.

III. Methods

A. Data Collection

Due to the individual features of a hogweed plant, large size of its leaves and large height, we found it possible to perform an aerial collection of the data. We have performed two flight missions in two separate days with the time difference of two weeks in order to cover crops in different growth stages, scanning the two neighboring fields in *x* city, *y* country,[2] and we used images captured from both fields separately for train and evaluation purposes. In both missions we collected RGB images of size 3000×4000 (height $\times$ width) pixels using DJI-FC330 camera mounted on a DJI drone capturing underneath field in spatial locations predefined during the mission planning. Specifically for the flight collecting training data, we have mounted an additional MAPIR Survey3 multispectral camera, capturing red (660 nm), green (550 nm) and NIR (850 nm) channels of the same spatial resolution at a predefined frequency of 0.5 Frames Per Second (FPS) and we denote the data from this camera as RGN.

In this research, we generate a fake infrared channel using a Generative Adversarial Neural Network. This approach is promising from a practical point of view. There are several alignment issues that we address using an artificial channel. 1) First of all, the camera DJI-FC330 is attached to the drone's gimbal. It allows a precise control of camera position during the flight according to the mission requirements. The gimbal is also used to stabilize the camera in case of wind gusts. On the other hand the multispectral camera is fixed and is always aligned with the drone's pose. Hence there is no way to compensate for its shift in case of any disturbances, and the resulted images from two cameras are not aligned with each other. 2) Due to different design of the lenses, the apertures of cameras are 20 centimeters away from each other, so both the field of view and the sensors illumination varies between the two. This effect also contributes to the

[1]A. Kuzmin, M. Nagel, S. Pitre, S. Pendyam, T. Blankevoort, and M. Welling "Taxonomy and evaluation ofstructured compression of convolutional neural networks". Available as an arXiv preprint at https://arxiv.org/abs/1912.09802.

[2]Location is hidden due to the blind review and the manuscript submission requirements.

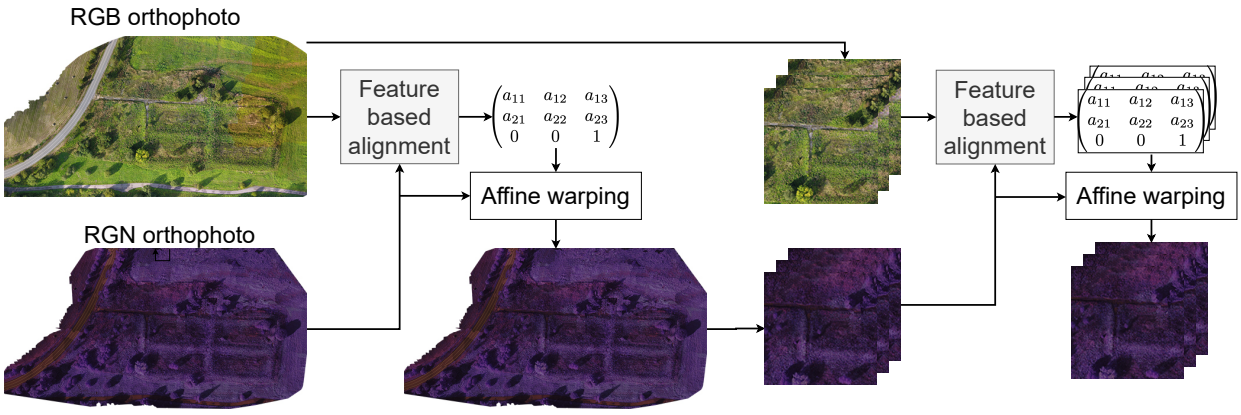


Fig. 1. Greedy pipeline of RGB and RGN orthophotos alignment and dataset creation for training neural networks. At first stage two orthophotos are aligned with a single affine transform, and the alignment is further refined by a second region-adaptive transform.

alignment problems. 3) Two cameras have different framerate and are not synchronised, so they do not perform every shot simultaneously. It raises the issue of alignment in case of real-time processing during the flight. 4) Using two cameras also lead to issues even in the case of two orthophotos alignment. There are separate GPS antennas for each camera. GPS data is critical for the orthophotos stitching, therefore it will influence the precision of the resulting orthophoto. 5) NIR cameras typically have lower resolution, then we need to crop images from RGB cameras in case of real-time data processing. It leads to the loss of data from the RGB camera, therefore decreasing the overall effectiveness of the system. 6) Finally, a multispectral camera is expensive in comparison to a single-board computer and a drone. Moreover, it is an additional payload. Then it will decrease the time of flight. To sum up, it is better to have a single cheap RGB camera on-board, rather than two cameras. To upgrade RGB camera with NIR capabilities, we can use the algorithm, described in this paper. As a result, the monitoring system with a single camera will be cheap and reliable.

B. Data Processing and Labeling

The collected dataset consists of two independent sequences of frames, which are not synchronized between each other. In order to align them, we propose to fuse both frames streams into a corresponding orthophotos, and we use an OpenDroneMap software for this [3]. Once the step is completed, it is convenient to label all the regions of interest within the RGB orthophoto, using any raster image processing software.

In order to align the images and construct a paired dataset for training, we have applied a greedy region-based procedure. At the first step, the two orthophotos are globally aligned based on the best affine transformation between corresponding ORB features [26]. At the second stage we divide both orthophotos and the labels into overlapping patches of size 256×256 px. We use the same alignment procedure applied to each patch independently and we search for the best match within the region of ± 100 pixels around the patch. We depict the overall pipeline in Fig. 1.

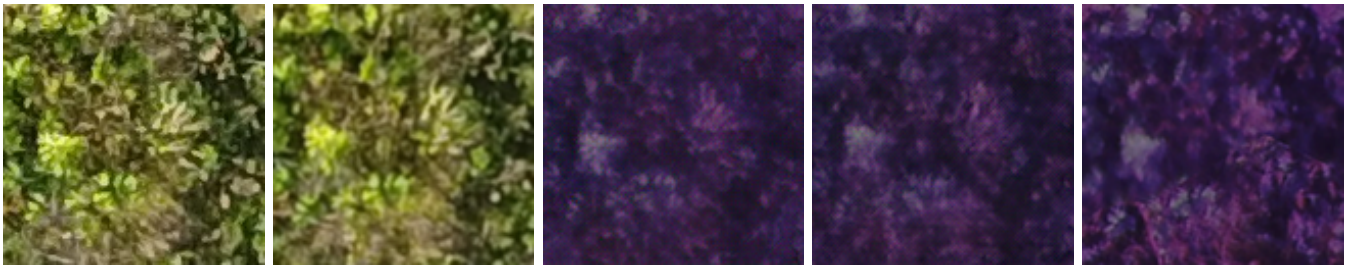
Constructing training data for the online scanning regime involves additional RGB frames stream used to create the orthophoto. For this scenario, an intermediate stage based on the same alignment procedure is required in order to find the position of the entire frame in the RGB orthophoto. The selected region is cropped from both orthophotos and labels. Due to the nature of texturing step used to synthesise orthophotos, the best match between original frame and its region inside the orthophoto is achieved within the central part of the frame. Thus we run the patch based refinement only on the central part of size 2000×2000 pixels. We have performed this procedure in order to sample both train and test data with random crops of size 128×128 for training purposes and 256×256 for the test. Since for the test dataset we do not collect multispectral images, we exclude this data from the discussed sampling procedure. Overall we have collected more than 20000 crops for training purposes and 83 crops for evaluation, and we carefully checked the correctness of the resulted labels by additional look at the images in the latter.

We should note, that the region, where performed data collection contained both vegetating (young) hogweed crops and those, which have already faded (old). Since the procedures required to eliminate these two types of hogweed plants can vary, we have labeled such cases with different semantic classed to learn the segmentation network to distinguish between them, and facilitating an effective practical application of our work. For all our reported data we have used a manual labelling procedure in GIMP graphic editor[4]. It should be noted that only RGB orthophotos are required for labelling, since the produced labels can be back-projected to the corresponding RGN orthophotos and original frames using procedure discussed above. We release the dataset consisting of both training, validation and testing samples, where we included both labeled orthophoto together with a paired crops suitable for training. Dataset is available at www.x.com.[5]

[3]ODM - A command line toolkit to generate maps, point clouds, 3D models and DEMs from drone, balloon or kite images. Available at https://github.com/OpenDroneMap/ODM.

[4]The GIMP Development Team. Available at https://www.gimp.org.

[5]It is the subject of double blind review.



RGB Frame | RGB Ortho | Fake NIR from frame | Fake NIR from ortho | Real NIR ortho

Fig. 2. Example of fake RGN frames synthesis with synthesis part of SynthFakeRGN model trained with frames and orthophoto data.

C. Neural Network Architectures

In this study, we do not provide extensive experiments with neural network architectures, since our main goal is to show the approaches to improve the performance of existing predefined architecture when additional data is at hand. However, we do compare different architectures in two modes, namely computationally more and less intensive, and we do the special emphasis on the latter one in this work due to practical concerns. We depart from the high capacity neural networks and use a SegNet architecture [27] with 29.5M parameters for segmentation part and an encoder-decoder network with residual blocks [28] and 7.8M parameters for the fake NIR synthesis. Both architectures were proven to perform well with the general-purpose segmentation and synthesis tasks respectively [29] [30]. Targeting a balance between the performance and inference speed when evaluated on an embedded system, we have selected well-known U-Net architectures for both segmentation and synthesis tasks in the less computationally intensive regime, since its superiority for satisfying both goals was shown in recent papers [7]. It is a fully Convolution Neural Network (CNN) which was initially developed specifically for semantic segmentation task, however it was also shown, that this network can efficiently solve image to image tasks [31]. Being an encoder-decoder architecture, the main benefit is that it allows processing and analysis of images at different scales, which allows to greatly increase its receptive field without the increase of computational complexity. Targeting the fast inference speeds, for segmentation network we have used a U-Net backbone consisting of 3 blocks in both encoder and decoder parts with 17.3M parameters total, and a backbone with 2 blocks and 2.2M parameters for synthesis. We found the capacity of these networks good enough to both obtain a good quality in semantic segmentation task, and run them with high speed on an embedded system.

D. Training and Implementation Details

We have conducted all our experiments using PyTorch framework [32], which allowed us to effectively deploy networks training, involving parallel computations, to GPUs. In all cases we have trained our networks using crops of size 128×128 pixels. We have used random crops as well as random multiple of 90° rotations and random horizontal and vertical flips to augment our training data. We minimised all objective functions using Adam optimizer [33] with learning rate of $1e^{-3}$, and a warm up scheduling was enabled for the first epoch. For training synthesis neural network, we have used an adversarial learning adopted from pix2pix [23]. Since the network chosen to perform synthesis is relatively small, we have used an adaptive gradient steps frequency [34] for both generator and discriminator in order to avoid discriminator's overfitting and other instabilities.

Overall, our training strategy is depicted in Fig. 3 and consists of three steps.

1) During the first one we train our network to segment hogweed plants based on both RGB and RGN data concatenated and passed as an input. For this task we perform a supervised learning by optimizing the cross-entropy objective, as it is shown in Fig. 3 a).
2) During the second step we train another neural network to synthesise fake RGN-like images out of the real RGB ones. The training of this stage is highlighted in Fig. 3 b). For this task we conduct the adversarial training similar to earlier proposed pix2pix model [23], and we use the same loss function of the form:

$$\begin{aligned} L_{\text{p2p}} = \mathbb{E}_{x,y} \log D(x, y, \theta_D) + \\ \mathbb{E}_{x} \log \left(1 - D(x, G(x, \theta_G), \theta_D)\right) + \\ \lambda_1 \, \mathbb{E}_{x,y} \, ||y - G(x, \theta_G)||_1. \end{aligned} \tag{1}$$

Here we denote RGB and RGN images as the x, y respectively, θ_G and θ_D as the trainable weights of generator and the discriminator networks G and D. Additionally, we involve the transfer learning and force our network to output images which should benefit to the target semantic segmentation task. For this purpose we penalise L2 objective L_{seg} computed between the output features of the network from the previous step evaluated on the ground-truth and fake RGN data:

$$L_{\text{seg}} = \mathbb{E}_{x,y} \, ||T(x, y, \theta_T) - T(x, G(x, \theta_G), \theta_T)||_2, \tag{2}$$

where we have denoted the segmentation network from the previous step and its weights as T and θ_T. We use the traditional Generative Adversarial Network (GAN) [35] min-max game to conduct the following optimization:

$$\theta_G = \arg \min_{\theta_G} \max_{\theta_D} L_{\text{p2p}} + \lambda_2 L_{\text{seg}}. \tag{3}$$

The intuition behind this step is to fuse information about valuable features from RGN images which can benefit segmentation of hogweed plants into the pre-processing network trained at this stage. We have found

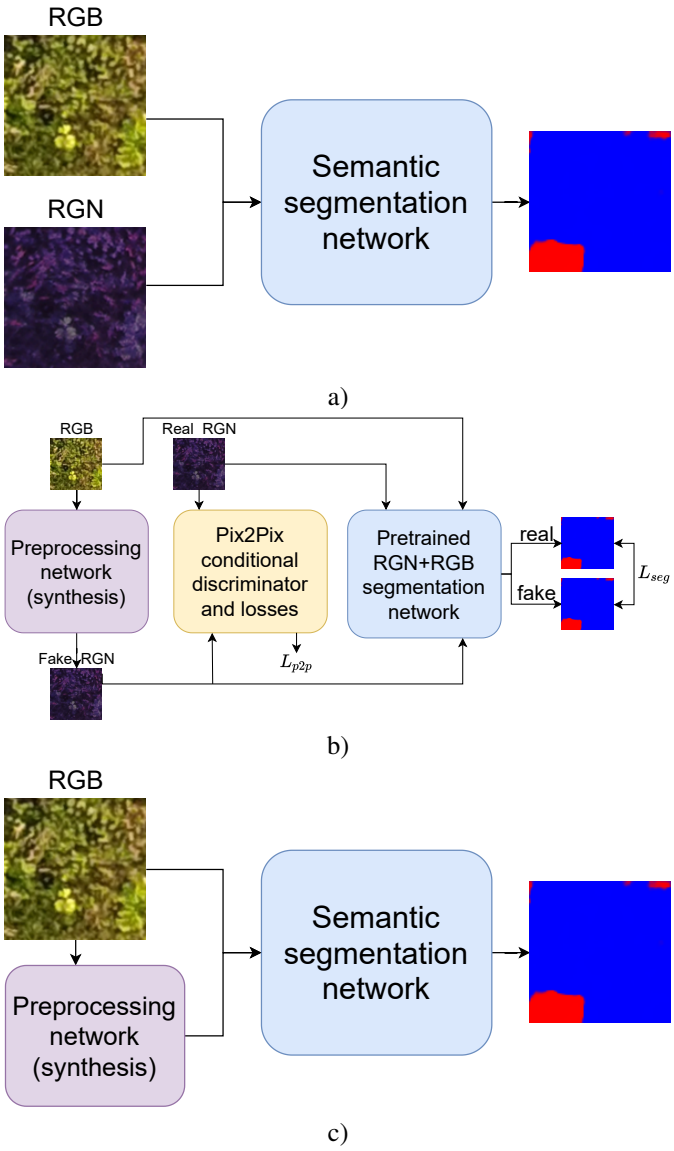


Fig. 3. Training procedure of the proposed segmentation pipeline: (a) the segmentation network is trained with real data, (b) the transfer learning procedure is used to train a synthesis network which provides the fake RGN frames, and (c) at the last stage the segmentation network is finetuned on the fake RGN data.

values $\lambda_1 = 100$ from Eq. (1) and $\lambda_2 = 10$ from Eq. (3) to suit well for this goal.

3) During the final step we perform an iterative inference of the network trained at the previous step through all our datasets samples to obtain fake RGN images and then finetune a semantic segmentation network on this data using the same procedure, as in the step 1. This step is depicted in Fig. 3 c) and does not require real RGN data anymore, since information about this domain is fused inside parameters of synthesis network during the previous stage.

E. Porting to Embedded System

We have used NVIDIA Jetson Xavier NX as a target device, since it was proven to be an efficient low-cost solution for various vision tasks in edge computing [8]. In order to accelerate the inference speed of our network and make simulations of the proposed approach running on an embedded system, we have applied the well-known techniques of network quantization and layers fusion (pruning). Due to specific choice of segmentation and synthesis networks in our less computationally intensive regime to be the standard U-Net architectures, the port of our models to the edge device is relatively straightforward. Due to this reason we have used NVIDIA TensorRT inference optimizer and runtime, which performs all the optimizations, as well as other architecture-specific tunings.

F. Evaluation Metrics

In order to evaluate our models and compare it to the baselines, we have used two most common approaches for the classification and semantic segmentation (i.e. classification of each image pixel), namely mean intersection over union (mIoU) and the Area Under the Curve (AUC) of a Receiver Operating Characteristic (ROC) curve, given by the true positive and false positive rates. For more informative evaluation both metrics are computed in a multi-class regime when young and old crops are treated as different classes and in a binary regime when both types of crops are treated as a single class. As a reference, we have observed, that typical values of 60 % of mIoU reported in various multi-class semantic segmentation benchmarks [36] is considered as an acceptable segmentation performance.

To assess the performance of our approach ported to embedded system, we measure the inference speed per 12MPx frame (3000×4000 px) in Frames Per Second (FPS), as well as the power consumption of the hardware in watts (W).

IV. Results

At the first stage of our study we validate the designed labels transfer. For this purpose we have randomly selected 10 frames out of the sequence used to construct the training dataset. The proposed strategy was used to sample images crops with corresponding labels transferred from orthophoto data. For the same set of frames we have performed manual labeling, which allowed to get the true labels for the sampled crops and use them to perform a quantitative comparison. The evaluation performed by the means of mIoU showed 95% and 90% for old and young crops respectively with the main source of error attributed to the inconsistencies introduced by independent manual labeling of frames and orthophoto. Such findings demonstrate that the difference between the true labels and the labels transferred from orthophoto with our proposed strategy is negligible. As it was extensively studied and proven in [37] [38], these small errors in labels do not affect the training and performance of neural networks. We also present an indicative example in Fig. 4 for a qualitative comparison.

We have trained our models separately with orthophoto images and with images from the camera. For both scenarios we trained the following models:

- Segmentation network working with RGB+RGN pair involving true RGN data (TrueRGN)

TABLE I

PERFORMANCE OF TRAINED LARGE MODELS (SEGNET+RESNET) EVALUATED ON TEST DATA FROM TRAIN AND TEST FIELDS. ROC-AUC AND MIOU METRICS ARE PRESENTED IN TABLE FOR MODELS TRAINED AND EVALUATED ON BOTH ORTHOPHOTO (ORTHO) AND FRAMES (FRAME) DATA. THE TYPE OF DATA USED FOR TRAINING IS GIVEN IN THE COLUMNS CAPTIONS UNDER THE NAME OF EACH MODEL, WHILE THE DATA TYPE USED FOR TESTING IS GIVEN IN THE CAPTION OF EACH ROW UNDER THE FIELD TYPE.

Dataset	Metric	TrueRGN		SynthFakeRGN		SynthFakeRGNnoTL		NoRGN		SynthNoRGN	
		Frame	Ortho	Frame	Ortho	Frame	Ortho	Frame	Ortho	Frame	Ortho
Test field, Ortho	Multi mIoU, %	N/A	N/A	**60.4**	58.1	55.9	53.7	56.8	54.2	54.5	53.4
	Binary mIoU, %			**82.4**	80.9	76.7	78.1	76.5	78.9	73.6	78.0
	Multi ROC-AUC			**0.921**	0.916	0.890	0.906	0.883	0.912	0.880	0.907
	Binary ROC-AUC			**0.905**	0.897	0.869	0.880	0.867	0.884	0.854	0.878
Test field, Frames	Multi mIoU, %	N/A	N/A	**57.5**	50.7	50.5	48.7	56.4	52.9	46.7	51.9
	Binary mIoU, %			**79.1**	73.2	69.1	71.7	75.1	77.7	63.7	76.3
	Multi ROC-AUC			**0.908**	0.885	0.862	0.879	0.884	0.905	0.840	0.899
	Binary ROC-AUC			**0.885**	0.852	0.823	0.841	0.858	0.877	0.792	0.868
Train field, Ortho	Multi mIoU, %	81.5	84.5	78.2	**85.1**	78.2	84.0	78.1	83.7	77.5	82.8
	Binary mIoU, %	82.2	**85.4**	79.3	85.3	79.1	84.7	79.7	84.3	78.8	83.0
	Multi ROC-AUC	0.924	0.938	0.909	**0.941**	0.909	0.935	0.910	0.934	0.906	0.930
	Binary ROC-AUC	0.915	**0.933**	0.901	0.932	0.900	0.929	0.907	0.927	0.899	0.920
Train field, Frames	Multi mIoU, %	83.3	81.0	**84.7**	62.7	81.3	67.6	82.0	68.4	81.9	69.6
	Binary mIoU, %	83.4	81.9	**84.8**	68.1	81.7	75.6	83.2	74.0	82.3	74.5
	Multi ROC-AUC	0.933	0.923	**0.939**	0.829	0.923	0.859	0.927	0.861	0.926	0.866
	Binary ROC-AUC	0.921	0.915	**0.930**	0.822	0.913	0.875	0.922	0.863	0.916	0.867

TABLE II

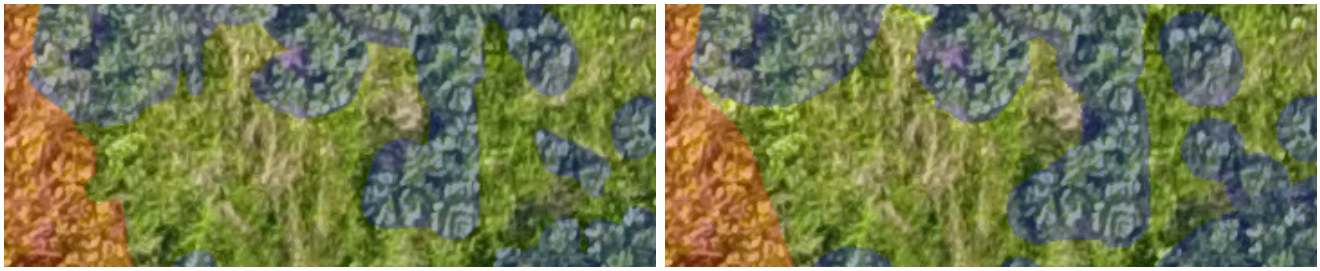
PERFORMANCE OF TRAINED SMALL MODELS (U-NETS) EVALUATED ON TEST DATA FROM TRAIN AND TEST FIELDS. ROC-AUC AND MIOU METRICS ARE PRESENTED IN TABLE FOR MODELS TRAINED AND EVALUATED ON BOTH ORTHOPHOTO (ORTHO) AND FRAMES (FRAME) DATA. THE TYPE OF DATA USED FOR TRAINING IS GIVEN IN THE COLUMNS CAPTIONS UNDER THE NAME OF EACH MODEL, WHILE THE DATA TYPE USED FOR TESTING IS GIVEN IN THE CAPTION OF EACH ROW UNDER THE FIELD TYPE.

Dataset	Metric	TrueRGN		SynthFakeRGN		SynthFakeRGNnoTL		NoRGN		SynthNoRGN	
		Frame	Ortho	Frame	Ortho	Frame	Ortho	Frame	Ortho	Frame	Ortho
Test field, Ortho	Multi mIoU, %	N/A	N/A	**65.1**	61.1	61.4	59.7	64.5	58.3	61.6	57.3
	Binary mIoU, %			**82.7**	80.0	82.0	79.6	81.2	76.4	79.1	72.2
	Multi ROC-AUC			**0.842**	0.818	0.820	0.810	0.829	0.789	0.823	0.784
	Binary ROC-AUC			**0.906**	0.891	0.902	0.889	0.895	0.865	0.884	0.837
Test field, Frames	Multi mIoU, %	N/A	N/A	**64.0**	55.9	60.7	54.2	59.8	54.5	59.5	52.9
	Binary mIoU, %			**80.8**	77.8	79.7	76.1	75.9	75.6	78.2	68.0
	Multi ROC-AUC			**0.838**	0.774	0.811	0.761	0.802	0.766	0.815	0.812
	Binary ROC-AUC			**0.893**	0.874	0.886	0.862	0.861	0.860	0.878	0.838
Train field, Ortho	Multi mIoU, %	75.7	81.6	72.7	**85.7**	68.2	83.7	73.9	81.5	69.4	79.9
	Binary mIoU, %	80.1	84.2	78.4	**88.5**	74.4	86.7	78.1	84.1	75.6	82.9
	Multi ROC-AUC	0.893	0.921	0.885	**0.943**	0.862	0.936	0.887	0.922	0.867	0.917
	Binary ROC-AUC	0.890	0.915	0.878	**0.939**	0.855	0.929	0.877	0.914	0.860	0.906
Train field, Frames	Multi mIoU, %	**78.8**	75.3	78.1	59.7	76.1	52.1	75.7	60.6	75.5	55.0
	Binary mIoU, %	81.6	78.3	**81.8**	72.5	80.3	71.4	79.4	71.7	79.2	64.4
	Multi ROC-AUC	0.909	0.892	**0.910**	0.810	0.900	0.794	0.895	0.812	0.895	0.770
	Binary ROC-AUC	**0.899**	0.879	**0.899**	0.843	0.891	0.833	0.886	0.838	0.884	0.790

- Segmentation network working with only RGB frames (NoRGN)
- Network consisting of synthesis and segmentation parts trained end-to-end with only RGB data (SynthNoRGN)
- Network consisting of synthesis and segmentation parts trained using transfer learning procedure with true RGN data (SynthFakeRGN).
- Network consisting of synthesis and segmentation parts trained without transfer learning procedure with true RGN data (SynthFakeRGNnoTL).

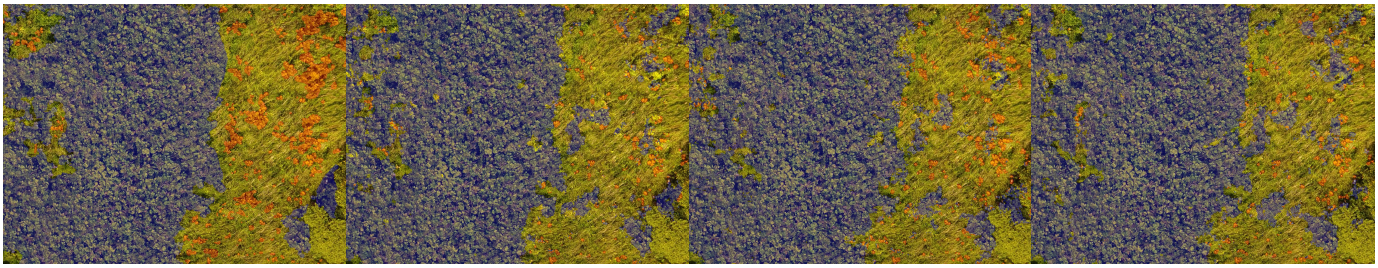
In Tabs. I and II we report on the results of all our models evaluated on both mission we performed directly for testing purposes (Test field) as well as on a subset of data obtained during collection of our training data (Train field). In Fig. 2 we report the visual example of synthesis U-Net network trained to mimic orthophoto data. It should be noted, that there is no intersection of training and testing data.

Several conclusions can be made from the tables. At first, metrics obtained on a subset of train field dataset are higher than those computed on test data. We think this change may indicate the difference in data collection and manual labeling performed for both train and test fields data separately at different time. The additional effect is overfitting, which is more pronounced for larger networks in Tab. I, where the gap between train and test fields is larger. Comparing the same models trained for either orthophoto data or frame data, but evaluated on both, it is clear that networks trained with frames are in general more robust with respect to the change of data type. Thus, the difference between multiclass mIoU metrics of SynthFakeRGN models trained with frames is on average 2% on test field data and 5.9% on train field one, while for the same models trained with orthophoto data it is 6.3% and 24.2% respectively. It means, that frames stream from the camera is more favourable for the purpose of hogweed precise monitoring, which supports our efforts in transferring all computations to the aerial platform.



True labels Transferred labels

Fig. 4. Qualitative comparison of true frame labels with those transferred from orthophoto. Old and young crops are denoted with red and blue colors respectively.



Target SynthFakeRGN SynthNoRGN NoRGN

Fig. 5. Performance of our models from Table I trained with orthophoto data on crop taken from the labeled othophoto of test field (best viewed in electronic version)

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TABLE III
PERFORMANCE OF TRAINED NETWORKS IN SIMULATIONS RAN ON AN EMBEDDED SYSTEM NVIDIA JETSON XAVIER NX AND ON SERVER GPU NVIDIA TESLA V100.

	NVIDIA Jetson Xavier NX			NVIDIA Tesla V100		
	TrueRGN/NoRGN	SynthFakeRGN/SynthNoRGN	SynthNet	TrueRGN/NoRGN	SynthFakeRGN/SynthNoRGN	SynthNet
Avg runtime, FPS	0.27	0.19	0.48	1.22	0.62	0.36
Avg power, W	6.5	6.5	.5	280	90	170

Comparing segmentation models trained with and without true RGN data we can emphasise the superiority of the former. Indeed, in all cases, when evaluated on data from train field, TrueRGN and SynthFakeRGN models constantly perform better than NoRGN and even its more computationally expensive analogue SynthNoRGN. For example, TrueRGN models trained and evaluated on frames data perform on average 2.2% better than corresponding NoRGN models and 2.4% better than SynthNoRGN ones in terms of multiclass mIoU metric. Also from both Tabs. I and II it is clear, that TrueRGN models better generalize to the different test data. As an example, U-Net model (Tab. II) trained on orthophoto data, gives score of 75.3 % when evaluated on frames data, which is 6.3 % less, than the same model evaluated on frames test set. At the same time, all other U-Net models trained on orhtophoto data overfit to the training domain which results in the differences of 26 %, 20.9 % and 24 % for SynthFakeRGN, NoRGN and SynthNoRGN respectively.

It is clearly seen from both tables that in almost all cases SynthFakeRGN model performs considerably better than NoRGN and SynthNoRGN with both mIoU and ROC-AUC metrics, which means, that it has benefited from training procedure introduced in this paper. For example, trained on orthophoto data, SynthFakeRGN models on average outperform NoRGN by 3.4 % and 0.017 in terms of multiclass mIoU and ROC-AUC metrics and by 2.8 % and 0.019 in terms of binary ones when evaluated on test dataset. It is also interesting to note, that being trained within the unified pipeline, SynthNoRGN performance in general deteriorates from a lighter NoRGN models, which have smaller number of parameters and computational complexity. This fact supports our approach, since it again highlights the importance of a proper training procedure.

Finally, we observe that the models utilising fake RGN data trained with transfer learning approach (SynthFakeRGN) continuously perform better their analogues trained with just RGB to NIR GAN (SynthFakeRGNnoTL). The highest difference occurs for models trained with frames data, where SynthFakeRGN outperform SynthFakeRGNnoTL by 5.1% on average in terms of multiclass mIoU metric measured for the test data. This fact proves, that the proposed transfer learning approach is a crucial component for fake NIR utilisation if semantic segmentation task is at hand.

Overall we observe, that multiclass metrics are lower than binary ones, and the difference is especially pronounced when evaluated on the test field. Indeed, the binary classification is an easier problem to solve, compared to the task of separating plants by age, however, even for the latter case, the metrics are relatively high, which shows the success of our approach. We present the visual comparison of multi-class segmenta-

tion performance between SynthFakeRGN, SynthNoRGN and NoRGN models with U-Net architecture in Fig. 5.

In order to show the direct applicability of our hogweed precise monitoring system, we have ported both U-Net semantic segmentation and fake NIR synthesis models to a Jetson Xavier NX embedded system and simulated its runtime during the mission by imitating a frames stream from RGB camera. Due to the memory limitations of this computational unit we have performed a tile inference, where each frame is firstly divided into non-overlapping regions of size 100×100 which are fed to the network with batch size 1, and the resulting local maps are stitched back to form a global map corresponding to the full frame resolution. Interestingly enough, we have managed to deploy all our networks to an embedded platform without any loss of segmentation quality, so we do not report the difference in metrics for networks evaluated on different platforms. In Table II we report on the runtime and a power consumption we have measured during the simulation. For comparison we also report values for the run on NVIDIA Tesla V100, where we have used our original models without any optimizations and we passed the whole 3000×4000 frame to the network.

V. Conclusions

In this article, we have proposed and evaluated an idea of incorporating knowledge about multispectral bands in the neural network performing semantic segmentation of Hogweed crops. We have collected and labeled two datasets consisting of old and young hogweed crops from two different fields within the same region, one of which involved only RGB frames stream and is used for testing purposes, and another one was collected with additional RGN camera on board. We have proposed an idea of labeling frames by labeling the corresponding orthophotos and we have successfully applied it for both datasets. We also proposed an alignment strategy, which allowed us to align RGB, RGN, and labels orthophotos between each other and RGB frames from the frame stream. With this data, we have trained a U-Net and SegNet architectures to perform semantic segmentation, and we have used a transfer learning procedure to train a separate synthesis head consisting of a another lighter network which performs the synthesis of RGN data to facilitate further semantic segmentation task. We have observed that this approach is at least 1.1 % better than end-to-end training of either segmentation or both synthesis and segmentation networks when only RGB data is involved. We have also shown, that this training procedure allows to obtain segmentation quality of the same level as when a real RGN data is provided. We have ported our trained networks to an embedded system and showed that they are capable of performing processing part of a real mission targeting precise location of hogweed crops. The proposed approach can be scaled to other agricultural industry related application including the diseases and insects detection.

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